

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Fuel thermal recirculation valve (TRV), Part Number 3691155
- Plug, Part Number 3089567
- Fuel System Clean Care Kit, Part Number 4919073.

Additional Service Items

- No additional service items required.

General Information

There are two variations of the fuel drain manifold:

1. Drain manifold TRV
2. Drain manifold TRV with check valve.

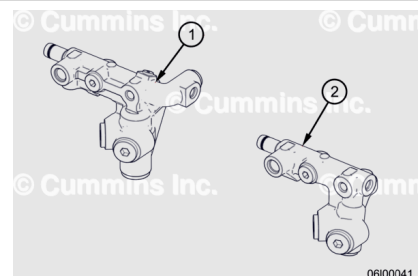
The TRV improves resistance to fuel filter gelling during engine operation in cold weather conditions.

Proper TRV operation requires all fuel supply and drain fittings to be air tight and properly tightened to specification.

Perform the steps in the following procedure before beginning this procedure to make sure all entrapped air is removed from the fuel system. Refer to Procedure 005-234 in Section 5.

(/qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html)

Note : The TRV is **not** controlled by the engine control module (ECM). A thermostat within the casting will contract when the fuel temperature is below 25°C [77°F], recirculating warm fuel return from various fuel system components into the cold fuel tank(s) and fuel supply plumbing. Similarly, this device will expand at temperatures above 25°C [77°F] containing the warm fuel supply, sending it back to the fuel tank(s).



Initial Check

Without Integrated Check Valve

Check the injector and high-pressure pump drain quick disconnect fittings (1).

Check the fuel pressure relief valve M12 connection (7).

Torque Value: 16 n•m [142 in-lb]

Check the fuel tank(s) supply M22 o-ring threaded fittings (2).

Torque Value: 47 n•m [35 ft-lb]

Check the TRV (TRV) thermostat M27 o-ring threaded plug (3).

Torque Value: 88 n•m [65 ft-lb]

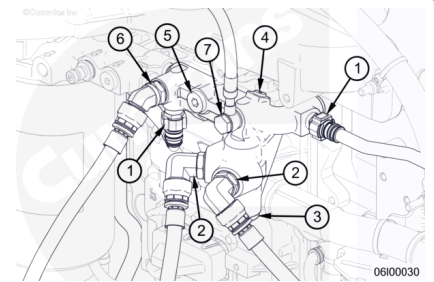
Check the TRV casting M8 o-ring threaded plug (4).

Torque Value: 5 n•m [44 in-lb]

Check TRV additional port M16 o-ring threaded plug (5).

Check the fuel tank drain M16 o-ring threaded fitting (6).

Torque Value: 27 n•m [20 in-lb]



With Integrated Check Valve

Check the injector and high-pressure pump drain quick disconnect fittings (1).

Check the fuel pressure relief valve M12 connection (6).

Torque Value: 16 n•m [142 in-lb]

Check the fuel tank(s) supply M22 o-ring threaded fittings (2).

Torque Value: 47 n•m [35 ft-lb]

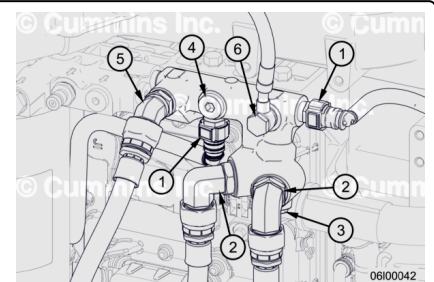
Check the TRV thermostat M27 o-ring threaded plug (3).

Torque Value: 47 n•m [35 ft-lb]

Check TRV additional port M16 o-ring threaded plug (4).

Check the fuel tank drain M16 o-ring threaded fitting (5).

Torque Value: 27 n•m [20 in-lb]



Flow Test

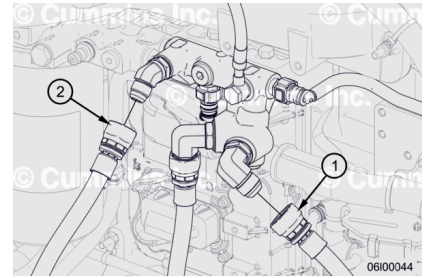
This test identifies potential air leaks limited to the supply or return fuel plumbing from Cummins Inc. or the original equipment manufacturer (OEM). Disconnect the OEM supply and return fuel lines at the TRV supply (1) and drain (2) ports.

Install a fuel supply line into the TRV supply (1) and drain (2) ports.

TRV drain manifold:

- Fuel supply line (1) - M22 fitting
- Fuel drain line (2) - M16 fitting

Install orifice diagnostic fuel line, Part Number 3164621, on pressure-side fuel filter outlet line.



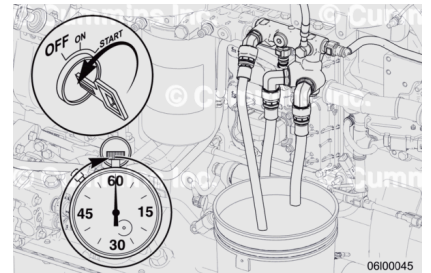
Make sure all connected supply lines are submerged in a bucket under clean fuel.

Turn keyswitch to ON position. Wait for the fuel lift pump cycle to complete (approximately 60 seconds). Cycle lift pump three times to eliminate air introduced through fuel line installations.

Start the engine and operate the engine speed to high idle for 60 seconds.

Observe bucket for excessive air bubbles exiting the installed fuel lines.

- If excessive air bubbling is identified, proceed to Leak Test in this procedure.
- If excessive air bubbling is **not** able to be identified, check OEM supply and return plumbing for proper torque specification. Contact OEM for further repair direction.

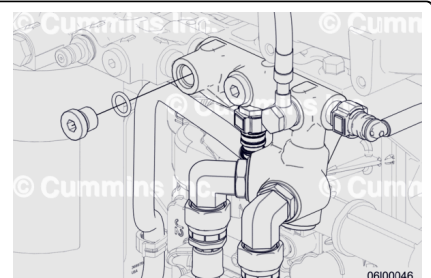


Leak Test

With Fuel TRV

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe



personal injury or death when working on the fuel system.

⚠ WARNING ⚠

Do not vent the fuel system on a hot engine, this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

Disconnect and cap the M16 x 1.5 OEM fuel return port fitting.
Use plug, Part Number 3089567.

Torque Value: 40 N·m [30 ft-lb]

Disconnect the right tank supply M22 x 1.5 fitting.

⚠ WARNING ⚠

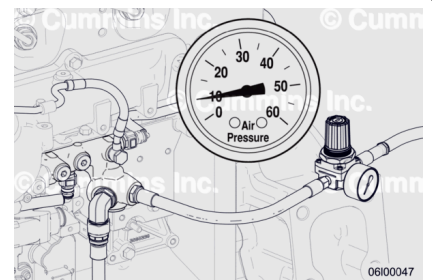
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Shop air pressure exceeding 207 kPa [30 psi] may cause damage to the fuel system.

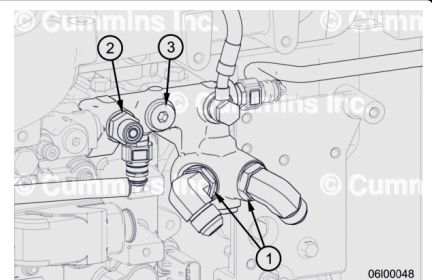
Connect compressed air pressure regulator assembly, Part Number 3164231, to M22 x 1.5 from tank fitting on fuel TRV.

- Pressurize the fuel system to 138 kPa [20 psi] and shut off the air supply.
- Close the air pressure regulator.



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- Watch for the pressure to decrease. The pressure **must not** decrease more than 69 kPa [10 psi] for 5 minutes.
- If the pressure decreases, use a spray bottle of soapy water to wet the pressure regulator assembly and threaded joints and Cummins® supply/return lines. Bubbles will appear if the connections or threaded joints are leaking.



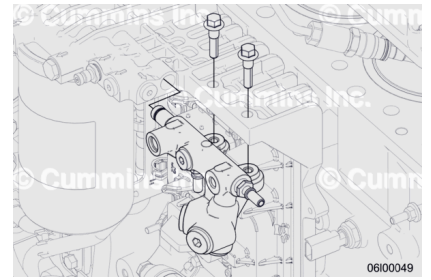
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Use a spray bottle of soapy water to help identify air leaks.

Check for air introduced through:

- Suction fuel lines

- OEM fuel lines
- Loose suction-side fuel filter assemblies, fittings, heating element, water in fuel sensor
- Fuel lift pump gasket or fittings
- Stand-pipe(s) in the fuel tank(s)
- TRV manifold inlet/outlet fittings
- Leaking fuel pump return line
- Leaking fuel rail return line
- Leaking fuel drain line from engine to the fuel tank.
- If leaks or other damage is found, replace the fuel system component. Follow the Install steps in the corresponding service manual.



Repair or replace any fuel system component found leaking or damaged. Follow installation instructions in the corresponding service manual.

Contact the OEM for guidance replacing OEM related components found leaking.

Note : If the pressure decreases and the hose connection(s) or threaded joints are not leaking, replace the TRV.

Preparatory Steps

⚠ WARNING ⚠

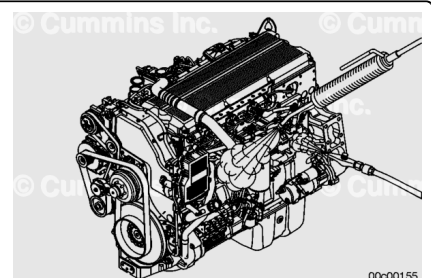
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.



Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.), which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

Clean the injector supply line connections and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

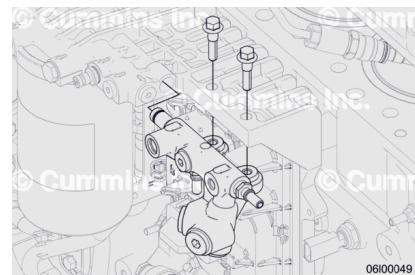
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the OEM fuel drain return line. See equipment manufacturer service information.
- Disconnect the fuel drain lines from the fuel drain manifold. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)

Remove

If applicable, remove the air intake manifold from the engine. Refer to Procedure 010-023 in Section 10.

(/qs3/pubsys2/xml/en/procedures/377/377-010-023.html)

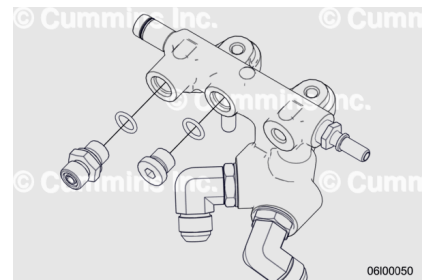
Remove the mounting capscrews and the thermal fuel recirculation valve.



Disassemble

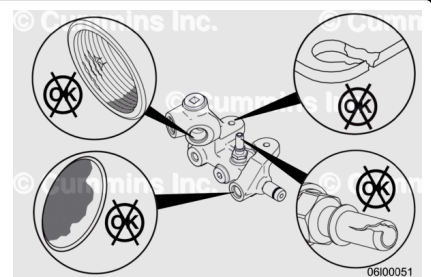
Remove the OEM fuel return fitting.

Remove the threaded plug.



Inspect for Reuse

Inspect the fuel drain manifold and for cracks, passage blockage, and material or debris.



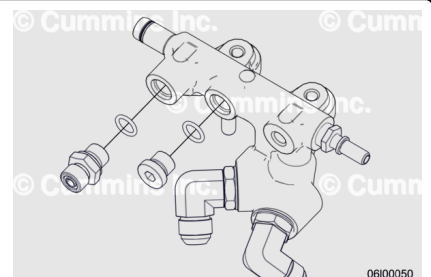
Assemble

Install the OEM fuel return fitting.

Torque Value: 40 n•m [30 ft-lb]

Install the threaded plug.

Torque Value: 37 n•m [27 ft-lb]

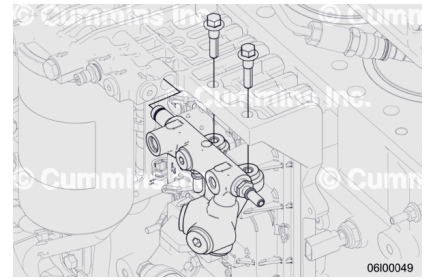


Install

Install the fuel drain manifold capscrews through the fuel filter head casting and into the fuel drain manifold.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the fuel supply lines.

- Connect the fuel supply line at the right side M22 x 1.5 port of the thermal recirculation manifold.
- Connect the fuel supply line at the left side M22 x 1.5 port of the TRV manifold.

Torque Value: 47 n•m [35 ft-lb]

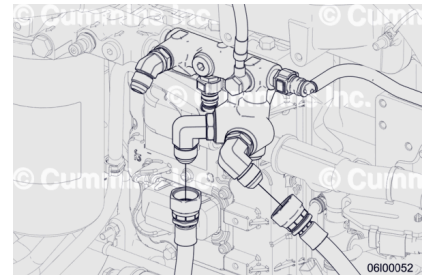
Refer to Procedure 200-001 in Section F.

(/qs3/pubsys2/xml/en/procedures/377/377-200-001.html)

If applicable, install the air intake manifold from the engine.

Refer to Procedure 010-023 in Section 10.

(/qs3/pubsys2/xml/en/procedures/377/377-010-023.html)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Bleeding air from the fuel system is an important step. Air in fuel will make an engine hard to start or no start.

- Connect the fuel drain lines. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)
- Connect the fuel lines. Make sure the connection gives an audible click and then give a slight pull on the fuel line to verify the connector is fully seated and locked.
- Connect the OEM fuel return line. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.

- Bleed air from the fuel system. Refer to Procedure 006-003 in Section 3.
(/qs3/pubsys2/xml/en/procedures/377/377-006-003.html)
- Operate the engine. Check for leaks.

Last Modified: 18-Nov-2021
